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Paper Code : CS-503

PUID : 05015 (To be mentioned in the main answer script)

DISCRETE MATHEMATICS

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

GROUP – A
(Multiple Choice Type Questions)

1. Choose the correct alternatives for any *ten* of the following : $10 \times 1 = 10$

   i) A 2-chromatic graph is

   a) a tree

   b) bipartite graph

   c) a cycle with odd number of vertices

   d) both (b) and (c).

   ii) The number of ways a null graph having 4 vertices can be properly coloured with 5 colours is

   a) 256 b) 1024

   c) 625 d) 125.

iii) The glb and lub of the set $\{(1, 2), \{1, 2, 4\}\}$ in the poset $\langle P(\{1, 2, 3, 4\}), \subseteq \rangle$ are

a) $\{2\}$ and $\{2, 4\}$ b) $\{\{1, 2\}$ and $\{1, 2, 4\}$

c) $\{1\}$ and $\{1, 2, 4\}$ d) $\{\varphi\}$ and $\{1, 2, 4\}$.

iv) A non-empty subset of N contains

a) maximal element b) minimal element

c) least element d) greatest element.

v) For any integer a, $\gcd(2a+1, 9a+4) =$

a) 2 b) 4

c) 1 d) 6.

vi) The number of ways an even sum is obtained when two indistinguishable dice are rolled is

a) 18 b) 12

c) 16 d) 14.

vii) The number 9420544 is divisible by

a) 36 b) 28

c) 24 d) none of these.

viii) The set of natural numbers N with the relationship '|' (divides) is a poset. How many minimal and maximal elements does it have ?

a) 1 minimal and 1 maximal

b) 1 minimal and 0 maximal

c) 1 minimal and more than 1 maximal

d) 0 minimal and 0 maximal.

ix) $A \wedge B$ is equivalent to which of the following ?

a) $\sim A \rightarrow \sim B$ b) $\sim A \rightarrow B$

c) $\sim B \rightarrow A$ d) $\sim (A \rightarrow \sim B)$.

x) Every integer is one of the forms

a) $7k, 2k \pm 1, 2k \pm 3$

b) $3k, 3k \pm 1, 3k \pm 2$

c) $4k \pm 1, 4k \pm 3$

d) none of these.

xi) Ramesh is paid on Friday in every week. Then in some month he is paid

a) twice b) thrice

c) four times d) none of these.

GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following. $3 \times 5 = 15$

2. Show that

$((p \vee q) \wedge \sim (\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$ is a tautology.

3. If $\gcd(a, b) = 1$, then show that.

$\gcd(a^2 - b^2, a^2 + b^2) = 1$ or 2

4. If we select 20 natural numbers, then show that at least 2 of them are such that their difference is divisible by 11.

5. Using characteristic root method solve the recurrence relation.

$a_n - 3a_{n-1} + 2a_{n-2} = n.3^n, n \geq 2$

with initial conditions $a_0 = 1,\ a_1 = 6.$

6. In a complemented distributive lattice, $< L, \wedge, \vee >$, prove that for

(i) $(a \wedge b)' = a' \vee b'$

(ii) $(a \vee b') = a' \wedge b'$

GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. a) Find the remainder when the sum

$1^5 + 2^5 + 3^5 + \ldots\ldots + 100^5$ is divided by 5.

b) For the poset $< S, | >$ where

$S = \{ 1, 2, 3, 4, 6, 9, 12, 18, 36 \}$ and '|' is the relation of divisibility defined on S. Draw the Hasse diagram. Find the minimal, maximal, greatest and least elements of the poset. Also find infimum and supremum for the sets $\{ 6, 18 \}$ and $\{ 4, 6, 9 \}$.

c) Solve the recurrence relation using generating function :

$a_n = 4(a_{n-1} - a_{n-2}) + 2^n,\ (n \geq 2);\ a_0 = 1,\ a_1 = 4.$

8. a) State generalized form of Pigeonhole Principle. Use it to find the total number of natural numbers that must be chosen to be sure of getting at least two of them whose difference is divisible by 11.

b) For the propositional variables p, q, r prove that the logical expression is tautology

$$(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$$

c) Use the theory of congruence to prove that $7 \mid (2^{5n+3} + 5^{2n+3})$, for $n \geq 1$.

9. a) Find the principle disjunctive normal form of each of the following statements :

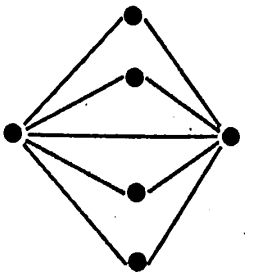
$$((\sim p \wedge q) \vee \sim (p \wedge r)) \wedge (p \vee (q \wedge r))$$

b) If $(L, \leq)$ be a lattice, then for any $a, b \in L$,

(i) $a \leq b \Leftrightarrow a \wedge b = a$

(ii) $a \leq b \Leftrightarrow a \vee b = b$

c) Find the chromatic polynomial for the following graph :



10. a) State the principle of Inclusion and Exclusion and use it to find the total number of integers between 1and 10000 which are neither perfect squares nor perfect cubes.

b) There are four processors p_1, p_2, p_3 and p_4 which are designated for six tasks t_1, t_2, t_3, t_4, t_5 and t_6 as follows :

$p_1 \rightarrow \{t_1, t_2\}$, $p_2 \rightarrow \{t_1, t_3, t_4\}$, $p_3 \rightarrow \{t_3, t_4, t_5\}$ and $p_4 \rightarrow \{t_2, t_6\}$. Is it possible to develop a super-processor which will serve exactly one task assigned to each of the four processors ? Solve this problem graphically and find a perfect match, if exists.

c) Find integers u and v satisfying :

$\gcd(272, 119) = 272u + 119v$.

11. a) Check the validity of the following arguments :

"If my program runs successfully then I will submit my project. I can appear in the examination only if I submit my project. Either my program runs successfully or the computer crashes. Therefore, if the computer does not crash then I can appear at the examination."

b) Solve : $y_{n+2} + 3y_{n+1} + 2y_n = 3^n$.

c) Define SDR of a family of finite sets. What is Hall's Marriage Condition ? Consider the family of finite sets $S = \{A_1, A_2, A_3, A_4\}$ where $A_1 = \{a, b, d, e\}$, $A_2 = \{b, c, d, e, f\}$, $A_3 = \{b, c, f\}$. Show whether S satisfies the marriage condition. If yes find a valid SDR of S.